Project Report

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Cyber Awareness Academy

An Anti-Corpo In-Browser Capture The Flag (CTF) Platform for Everyday Digital Literacy

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Abstract

Problem & Motivation: Most regular internet users know very little about cybersecurity even though their banking, private chats, and location history are all online. Because learning security usually involves Linux terminals, networking setups, and heavy coding, normal people feel intimidated and skip over it completely. Big tech companies take advantage of this gap to collect user data through hidden metadata, confusing terms of service, and fake encryption claims, which also leaves everyday users much more exposed when real hackers break into those companies.

Approach & Methodology: Using the Design Thinking process, our team ran a cybersecurity awareness survey with 130 college students and beginners ($n = 130$) to see why people struggle with digital privacy and quit existing learning websites. From this data, we built our Empathy Map and User Persona (Aditi Sharma), used the SCAMPER method to brainstorm features, and created *Cyber Awareness Academy*, an offline-capable, 100% in-browser Capture The Flag (CTF) web app built with React 19, a Node.js backend, and an auto-seeding SQLite3 database (`ctf.db`).

Key Results: In our survey of 130 respondents, 87 out of 130 felt cybersecurity requires basic technical literacy but is manageable, and 71 out of 130 felt genuine curiosity about it. However, 75 out of 130 rarely or never check their own digital footprint because 86 out of 130 feel existing tools assume an IT background and 29.5% do not know where to find reliable resources. When we tested our prototype with 22 beginners who had never played a CTF before, 100% passed our 3-step Flag Calibration tutorial on their first try, and 86.4% solved the warm-up questions and moved into the 5-stage linked investigation chains without touching a command line.

Contribution & Innovation: Our main contribution is combining a zero-download browser CTF across four live domains (OSINT, Steganography, Cryptography, and Web Security) with an anti-corporate story against a fictional monopoly called *EvilCorp* using our `CAT{...}` flag prefix, a slide-out textbook written in plain English, and a 2+5 connected question flow that helps beginners build real privacy habits.

1. Introduction

1.1 Context and Background

Almost everything we do today happens on a phone or a laptop. Whether we are paying for food over UPI, booking train tickets, or posting photos with friends, we leave behind a trail of personal information every single day. Even though people spend hours online, the average person knows almost nothing about cybersecurity. Most users treat their apps and websites like magic black boxes. You tap a button, the page loads, and as long as nothing crashes, you assume your data is safe.

When you ask normal people why they do not learn basic digital safety, the answer is almost always the same. Cybersecurity looks way too hard to get into. And honestly, it really does have a high skill ceiling. To someone outside computer science, learning security looks like an endless wall of Linux terminal commands, binary numbers, networking rules, and complicated code. Because people think you need to be an expert programmer just to start, they get put off and skip over even the simple stuff that affects their own accounts.

Big corporations and data brokers know this, and they use it to their advantage. Companies know a regular consumer is never going to check raw file headers, inspect browser cookies, or look at how their data is actually being sent over the network. Because nobody is checking, companies hide aggressive tracking behind friendly buttons. They bury unfair clauses at the bottom of long Terms of Service documents. Sometimes they even claim user telemetry is "encrypted" when it is just wrapped in basic Base64 encoding that anyone can reverse in two seconds. Even worse, when companies train users to blindly trust software while cutting corners on their own servers, like storing customer passwords as plain unsalted MD5 hashes, they make regular consumers much easier targets for real hackers. When a corporate database gets leaked, the company puts out a short apology, while the ordinary user gets their accounts hijacked because nobody ever showed them how password cracking or digital footprints work.

1.2 Literature Review and Market Survey

Capture The Flag (CTF) games have been used for years as a practical way to teach computer security. In a CTF, instead of just reading theory notes, players solve a real puzzle to find a hidden piece of text called a "flag" (in our platform, formatted like `CAT{secret_answer}`), which proves they actually found the flaw.

Carnegie Mellon University built **picoCTF** to introduce high school and college students to security puzzles using a game story. Community wargames like **OverTheWire** (especially the *Bandit* and *Natas* levels) have also helped thousands of students learn Linux shell commands and basic web security. More recently, big platforms like **TryHackMe** and **HackTheBox** have become popular by offering virtual labs where users can attack real virtual machines over a browser or VPN connection.

1.3 Research Gap and Problem Definition

When our team looked closely at the existing platforms during our background research, a big gap was obvious. Websites like HackTheBox and TryHackMe are great once you already know computer science basics, but they are way too confusing for a normal person or a first-year student who just wants to see how their privacy gets leaked online.

Specifically, existing platforms have four main problems for beginners:

1. **Too much setup trouble:** Most sites ask you to download OpenVPN files, boot up Linux virtual machines, or install command-line tools like `steghide`, `binwalk`, and `exiftool` before you can even try the first question.

2. **Jargon without simple explanations:** When a newcomer gets stuck on a traditional CTF, the site rarely explains the idea in normal words. The user has to dig through random forums full of confusing acronyms.

3. **Unrelated questions:** Most beginner CTFs give you random puzzles that have nothing to do with each other, so the player never sees how one small mistake in real life leads to a bigger compromise later.

4. **Cluttered interface design:** Many student projects copy generic, overly rounded web templates that look fake and crowd the screen instead of giving a clean place to work.

1.4 Scope, Objectives, and Contributions

The goal of our project, **Cyber Awareness Academy**, is to fix this problem by building a 100% in-browser CTF platform that needs zero setup and shows everyday people how corporate tracking and hacker tricks actually work. We built the whole platform around an "anti-corpo" story where the player acts as a rebel investigator exposing a shady monopoly called **EvilCorp**. Every flag on our platform uses the prefix `CAT{...}`.

Our main objectives and contributions are:

- **Zero local downloads:** Every question across all four arenas can be solved inside a normal web browser using free online tools like CyberChef, Photopea, HexEd.it, Google Lens, and browser DevTools.

- **The 2+5 connected question format:** Every domain has 7 questions. The first 2 questions are easy, unrelated warm-ups separated by a dotted line from the next 5 questions, which form a connected story chain where each answer helps unlock the next step.

- **Slide-Out Field Manual:** Every question card has a `[Theory]` button that opens a textbook chapter on the side of the screen, written in plain English with everyday analogies, real data breach stories, and step-by-step tool guides.

- **Offline-ready full-stack setup:** The project runs on React 19 for the frontend and Node.js with an embedded SQLite3 database (`ctf.db`) on the backend that automatically loads all 28 challenges as soon as you start the server.

1.5 Significance and Outline of the Report

We felt this project matters right now because corporate data collection and online scams are growing much faster than public awareness. Once a beginner solves a few easy puzzles on their own, they stop seeing computers as magic boxes and start paying attention to their own digital footprint, questioning fake security claims, and being more careful online.

The rest of this report follows the Design Thinking format required for our course. Section 2 covers our Methodology across the Empathy, Define, Ideation, and Prototyping stages, including our 130-student survey data, Aditi Sharma's persona and empathy map, SCAMPER analysis, system architecture, local setup steps, page-by-page features, and full writeups for all four CTF arenas. Section 3 wraps up the report with what our team learned, what we plan to improve next, and the open-source tools that inspired us, followed by our references.

2. Methodology (Design Thinking Framework)

To make sure we built something that genuinely helps beginners instead of just making another confusing hacker test, our team followed the five stages of **Design Thinking**: Empathy, Define, Ideation, Prototyping, and Testing.

2.1 Empathy Phase

2.1.1 Secondary Research (Market Survey)

Before writing any code, we tried out the four most popular websites people recommend for learning cybersecurity: **TryHackMe**, **HackTheBox**, **OverTheWire**, and **picoCTF**. We went through each one pretending we were a complete beginner with no Linux installation and zero terminal experience. Table 1 shows what we found.

Platform	**Target Audience**	**Strengths**	**Why Beginners Struggle**
TryHackMe	IT students & junior analysts	Guided rooms and browser-based attack boxes.	Paid subscriptions for best paths; heavy Linux & networking focus early on.
HackTheBox	Intermediate to advanced hackers	Realistic full-machine penetration testing labs.	Very steep entry barrier; requires OpenVPN and advanced CLI tools.
OverTheWire	CS students learning Linux	Free, lightweight, great for learning bash commands.	Pure SSH terminal text; zero visual interface or beginner analogies.
picoCTF	High school & college students	Free, fun puzzles across crypto, stego, and web.	Many challenges still require local Python scripts or Linux binary tools.
Cyber Awareness Academy (Ours)	**Complete beginners & everyday users**	**100% in-browser, zero local installs, built-in plain-English textbook, 2+5 linked story chain.**	Currently runs offline locally (server hosting planned next).

Table 1. Market comparison of existing cybersecurity learning platforms.

2.1.2 Primary Research (Survey Form and Interviews)

To back up our market check with real numbers from people around us, we ran a Cybersecurity Awareness Survey that gathered **130 responses** (n = 130), mostly from college students aged 19 to 22 who are curious about tech but come from non-technical or beginner backgrounds. Going through the survey charts and written responses gave us a really clear picture of why people hold back from learning digital privacy:

- **Biggest Hurdle to Maintaining Personal Digital Privacy (129 responses):** When we asked people what single thing stops them the most from keeping their digital privacy safe, the number one answer was *"I do not know where to find reliable resources"* at **29.5% (38 out of 130)**. Second was *"Convenience over security"* at **16.3%**. Tied right behind that at **14.7%** each were *"The terminology is too technical and confusing"* and *"Too many accounts to manage."* Another **11.6%** picked *"I do not believe I am a target so it feels unnecessary"* (often thinking their personal data is not that special), and **9.3%** said *"It requires too much time to learn."*

- **Trade-Off Between Security and Convenience (127 responses):** When we asked how they feel about the daily balance between security and convenience, **37%** felt the current balance is fair, and **34.6%** said they are willing to accept serious inconvenience if it actually guarantees security. At the same time, **22.8% (29 out of 130)** admitted that security measures feel annoying even though they put up with them, while the remaining **~5.6%** saw security steps as an unnecessary roadblock.

- **How Beginners Think and Feel About Cybersecurity:** Most respondents (**87 out of 130**) said they believe cybersecurity *"requires basic technical literacy, but is manageable"* in principle, and **71 out of 130** said their most honest reaction to the subject is simple **curiosity**. However, **86 out of 130 (around 64%)** assumed that existing cybersecurity tools and tutorials are built for people who already have an IT or programming background. Because of that, roughly **one-third (~1/3)** reported feeling overwhelmed, anxious, confused, or bored whenever they try to read about it.

- **What Beginners Actually Do Online:** While **107 out of 130** could rightly spot `HTTPS` as the secure web protocol, **75 out of 130** admitted they rarely or never check their own digital footprint. Many students also told us they avoid asking beginner questions on tech forums because they are scared of being judged or made to feel "stupid."

In the open text responses, people kept asking for the same thing: drop the heavy acronyms. Multiple students told us *"Simplifying jargon is definitely a start,"* and one response summed up the whole problem in one line: *"Explain it to me like I'm 5 years old."*

2.1.3 Stakeholder Canvas

From our survey findings, we mapped out the three groups of stakeholders involved in our project:

- **Primary Stakeholders (Direct Users):** Everyday internet users, college students aged 19 to 22, and curious beginners (represented by our persona Aditi Sharma) who want to understand digital privacy without needing an IT degree first.

- **Secondary Stakeholders (Educators & Mentors):** College faculty, student club leads, and workshop hosts who need a zero-setup browser platform to teach cybersecurity concepts in class without spending hours fixing virtual machines on student laptops.

- **External Stakeholders:** Creators of free browser utilities (such as GCHQ CyberChef and Photopea), privacy educators, and regular consumers whose data is collected by tech platforms.

2.1.4 Empathy Map and User Persona

We combined the data and quotes from our 130-respondent survey into a six-section Empathy Map (*Says, Thinks, Feels, Does, Pains,* and *Gains*), shown in Fig. 1, and created our main User Persona, **Aditi Sharma**.

User Persona: Aditi Sharma

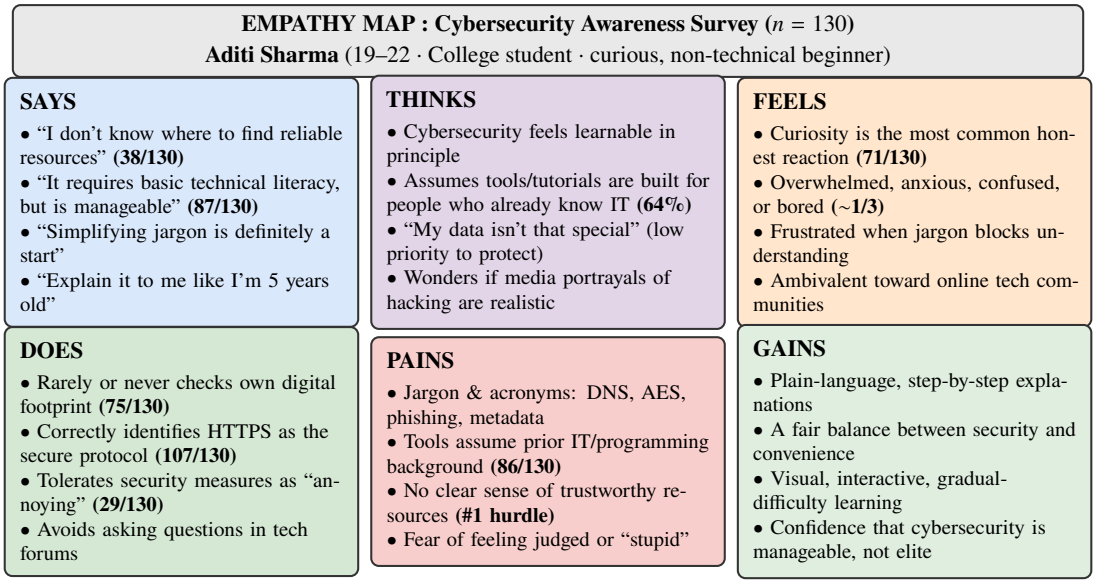


Fig. 1. Empathy map for Aditi Sharma based on the Cybersecurity Awareness Survey ($n = 130$).

- **The Curious Beginner:** Aditi is a 19-year-old college student who represents the non-technical beginner dealing with everyday digital privacy decisions. Aditi genuinely wants to learn how security works, and she knows basic things like checking for `HTTPS`, but she gets immediately intimidated by heavy technical jargon like `DNS`, `AES`, metadata, or phishing whenever she opens a normal tutorial.

- **The Say-Do Gap:** Aditi claims curiosity (like 71 out of 130 students in our survey) and says security matters, but she rarely checks her own digital footprint in practice (75 out of 130). The main barrier is that she does not know where to find reliable resources (29.5%). On top of that, she fears feeling judged in online tech communities, so she avoids asking questions and stays exposed to corporate data mining.

2.2 Define Phase

In the Define phase, our team sat together and looked at what the survey and Aditi's persona were actually telling us. At the start of the semester, our rough idea was just: *"People do not know cybersecurity because it is hard, so let's make an easy CTF website."*

Once we saw the "Say-Do Gap" in our 130 survey responses, we realized that just making puzzles easier was not enough. People are already curious, but they stop because they do not know where to find trustworthy beginner resources, they hate unexplained jargon, and they do not want to install complicated IT tools. We narrowed this down into our final **Problem Statement**:

> *"Curious, non-technical beginners like Aditi want to understand digital privacy, yet more than half never check their own digital footprint because they lack a single trustworthy beginner resource, feel locked out by unexplained jargon like DNS or AES, and assume security tools require an IT background. This gap lets corporations quietly mine consumer data and leaves everyday users vulnerable to attacks. How might we build a friendly, zero-installation, 100% in-browser CTF platform that*

pairs plain-English step-by-step explanations with visual, gradual-difficulty puzzles so beginners can build real confidence without feeling judged?"

2.3 Ideation Phase

2.3.1 Brainstorming and Mind Mapping

Once we had a clear problem statement, the four of us brainstormed how to build a platform that directly solves Aditi's pains and delivers her gains. We sketched out a Mind Map (shown in Fig. 2) split into four main parts: Anti-Corpo Story, 2+5 Gradual Flow, Zero-Install Learning, and Terminal UI.

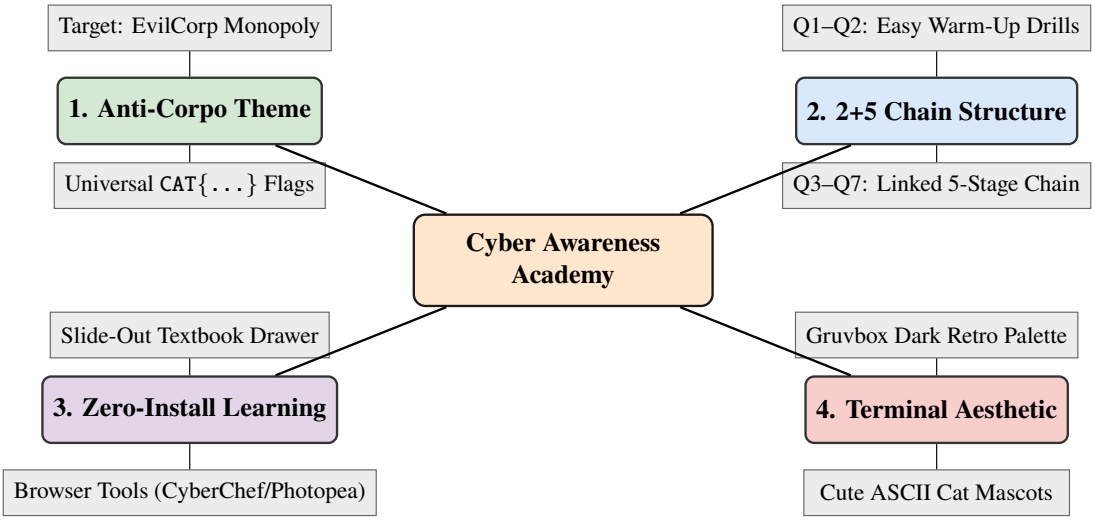


Fig. 2. Ideation mind map organizing the four main parts of Cyber Awareness Academy.

2.3.2 Application of the SCAMPER Method

We also used the **SCAMPER** method to rethink how a normal CTF works and make it fit a beginner like Aditi:

- **Substitute:** We replaced local Linux command-line tools (like `steghide`, `xxd`, and `john`) with free, browser-based tools (CyberChef, Photopea, HexEd.it, CrackStation, and Academo) so nobody has to install anything.
- **Combine:** We combined the CTF challenge cards with a slide-out textbook panel (`TextbookPanel.js`) on the exact same page, including a "Jargon Buster" box that explains terms like DNS, AES, EXIF, and SQL in plain words.
- **Adapt:** We adapted real-world privacy leaks (like the 2012 LinkedIn unsalted hash breach and the Strava fitness heatmap leak) into our story about exposing *EvilCorp*.
- **Modify:** We changed the usual flat list of puzzles into our **2+5 format** (two easy warm-ups, a dotted line divider, and five connected questions that get slightly trickier step by step). We also modified the design into a clean Gruvbox Dark terminal look with friendly ASCII cats.

- **Put to another use:** We used everyday websites that students already know (Instagram search, YouTube community posts, Google Maps library timings, and Google Lens) as investigation tools in our OSINT arena.
- **Eliminate:** We got rid of virtual machines, OpenVPN files, sign-up forms, and complicated database setup by using a self-seeding SQLite file (`ctf.db`) and a forgiving answer checker.
- **Reverse:** Instead of asking the student to defend a company like boring corporate training videos do, we flipped the role: the player investigates how *EvilCorp* abuses user trust.

2.4 Prototyping Phase

2.4.1 Low-Fidelity Prototype and UI Evolution

When we made our first prototype in React, we fell into a trap that a lot of student web projects fall into. Our early version had big rounded borders around every tiny piece of text, glowing neon outlines, and three fake macOS window dots in the corner of every box. When we looked at it honestly, it looked artificial and messy. Also, in our earliest build, we only had one domain working while the others were just placeholder cards.

We decided to clean up the whole design and build out all four domains properly:

1. We removed the fake macOS window dots and extra wrapper boxes so that borders are only used for actual question cards and briefings.

2. We switched to sharp rectangular terminal corners using the **Gruvbox Dark** color scheme (`index.css` and `App.css`) and `Fira Code` monospace text.

3. We added cute ASCII cat companions (`ascii.js`) so the dark screen still feels welcoming and fun for someone who is nervous about tech.

4. We built all four domains into complete 7-question arenas (`Challenges.js`, `StegChallenges.js`, `CryptoChallenges.js`, and `WebSecChallenges.js`), including downloadable evidence files in `/public/steg/` and seven live EvilCorp webpages in `/public/websec/`.

2.4.2 Final Prototype Architecture and Tech Stack

Our final application uses a **React 19** frontend (running on port `3000`) talking to a **Node.js HTTP REST API** backend (`server.js` on port `3001`) backed by an embedded **SQLite3** database (`ctf.db`). Fig. 3 shows how user answers flow through the system and update the interface.

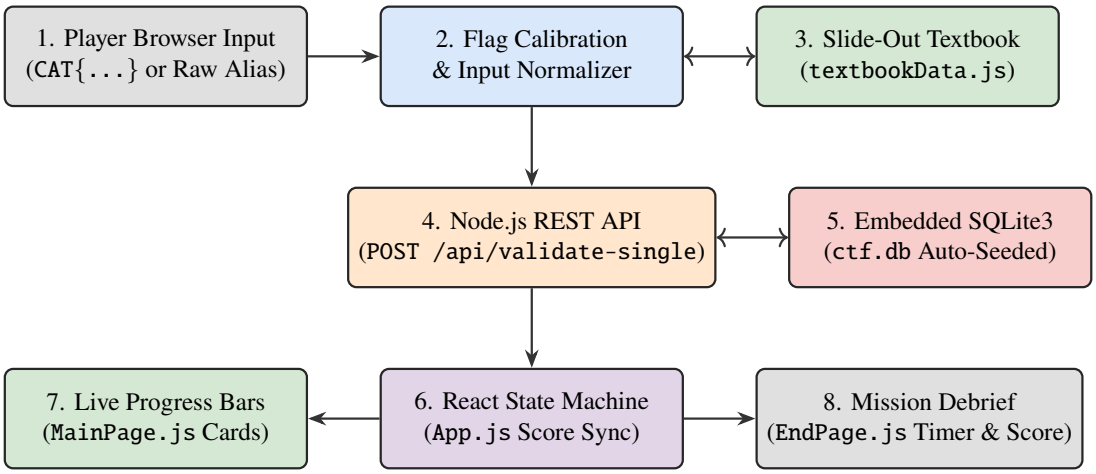


Fig. 3. Cyber Awareness Academy full-stack answer checking and score update flow.

When a player types an answer and hits `Enter` or clicks `[Submit]`, our input checker trims extra spaces, ignores capital letters, and accepts both `CAT{...}` and raw answers so beginners never fail just because of a bracket typo. At the same time, it sends a `POST` request to

`/api/validate-single` on port `3001`, where `server.js` checks the `ctf_answers` table in `ctf.db`. Once a question is marked solved, `App.js` immediately updates that domain's score so the progress bar on the main dashboard stays accurate.

2.4.3 Offline Installation and Quickstart Guide (From GitHub README)

Right now, our project runs as an offline-capable local setup so anyone can clone it and test it on their own machine without needing cloud keys or internet hosting. Later, we plan to put it on our own Linux server. As written in our GitHub `README.md`, running the app locally takes four quick steps:

1. **Prerequisites:** Install Node.js (v16 or newer) from https://nodejs.org.
2. **Clone and Install:** Clone the repository and install the npm packages:

```
PowerShell / Terminal ::  Step 1 & 2
$ git clone https://github.com/Aweek07/CTF_PROJECT.git
$ cd CTF_PROJECT
$ npm install
```

3. **Start the SQLite Backend Server (Terminal 1):** Run `node server.js`. It starts on port `3001` and automatically creates and seeds `ctf.db` with all 28 domain flags:

```
Terminal 1 ::  Backend API & SQLite Seeding
$ node server.js
SQLite database initialized with Anti-Corpo CAT{...} challenges.
Backend server running on http://localhost:3001
```

4. **Launch the React Frontend (Terminal 2):** Open a second terminal and run `npm start` to open the app in your browser at `http://localhost:3000`:

```
Terminal 2 ::  React Gruvbox Terminal UI
$ npm start
Compiled successfully!  Local:  http://localhost:3000
```

2.4.4 Page-by-Page Walkthrough and Platform Features

We designed every screen in the app to make sure a beginner like Aditi feels guided from the moment she opens the site:

1. **Terminal Navigation Bar (`NavBar.js`):** A top bar that shows a live terminal path matching whichever page the player is on (such as `guest@cyber-academy:~/dashboard/websec/arena$`). Inside an arena, it shows a live `[SOLVED: X/7]` counter and a timer. We also connected `App.js` to `window.history.pushState` so the normal browser Back and Forward buttons work smoothly between pages.

2. **The Anti-Corpo Manifesto (`HomePage.js`):** The home screen welcomes the user with an ASCII banner and a waving cat mascot. It explains our mission in three short blocks: *01. The Illusion of Digital Safety* (how big companies take advantage of regular users), *02. What is a Capture The Flag (CTF)?*, and *03. Our Aim: Cyber Awareness for Everyone*.

3. **Guided Flag Calibration Tutorial (`Onboarding.js`):** Before sending a new user into the main challenges, this 3-step practice page lets them test how flags work:

   - *Drill 1:* Entering `CAT{welcome_to_cybersec}` to see how the `CAT{...}` brackets work.
   - *Drill 2:* Entering `CAT{look_beneath_the_surface}` to practice underscores.
   - *Drill 3:* Entering `CAT{ready_to_hack}` to unlock the main dashboard.

   It even has an `[Auto-fill]` button so nobody gets stuck on the tutorial.

4. **Cybersecurity Domains Dashboard (`MainPage.js`):** Shows the four main arena cards (**OSINT**, **Steganography**, **Cryptography**, and **Web Security**). Each card displays custom ASCII art, the free browser tools used inside, the topics covered, and a live progress bar (`Solved: X/7`).

5. **Discipline Mission Dossiers (`DomainPage.js`):** Clicking any domain card opens a short mission briefing before the questions begin. It explains what the topic means in simple terms, how professionals use it, a red `[!] Threat Profile` box showing how regular people get affected by it, and the tools needed.

6. **The Slide-Out Field Manual (`TextbookPanel.js` & `textbookData.js`):** Because our survey showed that 29.5% of beginners do not know where to find reliable resources and hate heavy jargon, every single question card has a purple `[Theory]` button. Clicking it slides open a chapter written specifically for that question (28 chapters in total). Each chapter explains the concept in plain language, gives a simple everyday analogy, shares a real-world story, lists step-by-step instructions, and includes a Jargon Buster dictionary.

7. **Mission Accomplished Debrief (`EndPage.js`):** Once a user finishes an arena, this page rewards them with an ASCII trophy cat, shows their final time and score out of 7, recaps what they learned, and lists our team credits.

2.4.5 Comprehensive CTF Writeups and Learning Breakdown

Every arena in our project follows the same **2+5 layout** (shown in Fig. 4). Questions 1 and 2 are easy, unrelated warm-ups so a beginner can get two quick wins and understand the flow. Below a dotted divider line, Questions 3 to 7 are all connected in a 5-stage story chain where solving one question gives you the clue, password, or ID needed for the next one.

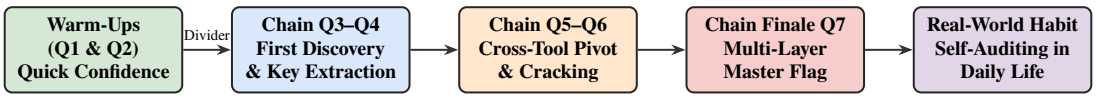


Fig. 4. The 2+5 challenge flow from warm-up drills to connected investigation stages.

Below is the step-by-step writeup for all four arenas and how each puzzle helps a beginner learn.

A. Open Source Intelligence (OSINT) Arena (`Challenges.js`) The OSINT arena shows players how public details left on maps, network records, Instagram, and YouTube can be pieced together to track a real person's routine.

- **Q1: Coordinates (Warm-Up 1):** The question gives the coordinates `48.8584, 2.2945` and asks what famous landmark is there. Pasting those numbers into Google Maps drops a pin right on the Eiffel Tower in Paris, giving the flag **`CAT`**{**`eiffeltower`**}. *How it helps beginners:* It takes ten seconds to solve and makes players realize that phone cameras save these exact GPS numbers inside photo files.

- **Q2: Who Owns This IP? (Warm-Up 2):** Players are asked which company runs the IP address `8.8.8.8`. Searching `8.8.8.8` on `ipinfo.io` shows it belongs to Google (`AS15169`), giving **`CAT`**{**`google`**}. *How it helps beginners:* It shows that IP addresses are registered to real companies and explains what a DNS server actually does when you type a website name.

- *Dotted Line Divider (Start of 5-Stage Connected OSINT Investigation)*

- **Q3: The Target Profile (Chain Stage 1):** Players search Instagram for "a guy who hates to click photos" and check what sport he follows. Finding the account and looking at his posts and highlights shows football matches, giving **`CAT`**{**`footbal`**}. *How it helps beginners:* Even someone who never posts their own face still reveals personal interests that scammers use to start fake conversations.

- **Q4: Pattern of Life (Chain Stage 2):** Checking the location tags across his Instagram posts shows one Mumbai area popping up again and again: Churchgate (**`CAT`**{**`church_gate`**}). *How it helps beginners:* Tagging a place once shows where you visited, but tagging places every week maps out your daily routine, just like the Strava army base heatmap leak we explain in the theory tab.

- **Q5: The Start (Chain Stage 3):** The question asks for the location of his very first Instagram post, which has no location tag. Taking a screenshot of the photo and searching it with **Google Lens** matches the painted domes to the Beatles Ashram in Rishikesh,

giving **CAT**{**beatlesashram**}. *How it helps beginners:* It proves that deleting a location tag does not hide where you are if landmarks are visible behind you.

- **Q6: The Connection (Chain Stage 4):** Players need to find the target's real first name. Searching his Instagram username across Google and YouTube turns up a YouTube community post tied to the same handle that mentions his first name, Abhyudaya (**CAT**{**abhyudaya**}). *How it helps beginners:* Almost everyone reuses the same username across apps, so mentioning your real name on one site connects back to all your other accounts.

- **Q7: The Final Clue (Chain Stage 5):** Players check the newest photo on the account, figure out the neighborhood, find the public library in that area on Google Maps, and look up its Monday timings in 24-hour format to get **CAT**{**9:30-19:00**}. *How it helps beginners:* It shows how investigators combine social media posts with public map listings to confirm real-world timings.

B. Steganography Arena (`StegChallenges.js`) The Steganography arena follows an insider sneaking secret files out of *EvilCorp*. Every question has a downloadable file in `/public/steg/`.

- **Q1: Evil Terms of Service (Warm-Up 1):** Players download `Evil_Terms_Of_Service.docx`, which claims you sign away your rights by scrolling to the bottom. Pressing `Ctrl+A` to highlight all text and changing the font color reveals white text hidden on the white page: **CAT**{**but_i_refuse**}. *How it helps beginners:* It shows how documents can hide invisible text right in front of you.

- **Q2: Evil Mug (Warm-Up 2):** Players download `evil_mug.jpg`, a normal picture of a coffee mug. Checking the file's EXIF details (using Windows `Properties -> Details` or `exif.tools`) shows a hidden note in the `Comments` tag: **CAT**{**detailed_search**}. *How it helps beginners:* It shows that photos carry hidden text tags like camera model, date, and comments along with the picture.

- *Dotted Line Divider (Start of 5-Stage Whistleblower Chain)*

- **Q3: BLACKBOOK (Chain Stage 1):** The file `blackbook.png` looks like a plain black square. Opening it in **Photopea** (`photopea.com`) and dragging the Brightness/Levels slider up lights up dark grey letters (`RGB(4,4,4)` on `RGB(0,0,0)`) spelling **CAT**{**l1ghtw0rks**}. *How it helps beginners:* It teaches why blacking out sensitive info in a screenshot with a semi-transparent brush is dangerous because anyone can brighten the image to read what was underneath.

- **Q4: Gimmick Boss (Chain Stage 2):** In `gimmick_boss.jpg`, checking the EXIF comment first gives `password:lifeisajoke`. Entering `lifeisajoke` into an online **Steghide** extractor pulls out a hidden text file ending with 8-bit binary numbers. Converting that binary to text gives **CAT**{**whysoserious**}. *How it helps beginners:* Players see how

password-protected files can be tucked inside image pixels and how binary numbers turn into letters.

- **Q5: EvilChair (Chain Stage 3):** The photo `evilchair.jpg` shows an office chair, but opening it in the browser hex editor **HexEd.it** and scrolling to the bottom shows plain text written right after the JPEG end marker (`FF D9`): **CAT**{**saveaseat**}. *How it helps beginners:* Photo apps stop reading a JPEG once they hit `FF D9`, so extra data pasted at the very end stays invisible in normal gallery apps.

- **Q6: EvilCorp Memory Wipe (Chain Stage 4):** Players download the audio file `memory_wipe.wav`. Following the hint to *look* at the sound instead of just listening, uploading it to an online **Spectrogram Viewer** (Academo) shows the flag drawn inside the sound waves between 5 kHz and 15 kHz: **cat**{**waverunner**}. *How it helps beginners:* Seeing words drawn inside sound frequencies is a really cool moment that gets beginners interested in audio forensics.

- **Q7: tryfindme (Chain Stage 5):** The last file, `tryfindme.jpg`, has a metadata note saying the password is in the photo itself. Brightening the photo in Photopea reveals the number `0004`. Using `0004` as the Steghide password extracts a text file containing `Q0FUe3lvdWZvdW5kbWUhfQ==`, which decodes from Base64 into **CAT**{**youfoundme!**}. *How it helps beginners:* It brings together metadata checking, brightness adjustment, Steghide extraction, and Base64 decoding in one final puzzle.

C. Cryptography Arena (`CryptoChallenges.js`) The Cryptography arena (*Operation Blackbox*) shows how companies throw around the word "encrypted" while using weak encoding and unsalted password hashes.

- **Q1: The Leaked Executive Memo (Warm-Up 1):** EvilCorp shifted a memo by 13 letters: `PNG{rivypbec_cevinpl_vf_n_zlgu}`. Using **ROT13** in CyberChef shifts every letter back 13 spots to get **CAT**{**evilcorp_privacy_is_a_myth**}. *How it helps beginners:* Shows how letter-shifting ciphers work and why they are easy to reverse.

- **Q2: The Fake Telemetry "Encryption" (Warm-Up 2):** EvilCorp claims user tracking data is encrypted, but the string ends with `==`: `Q0FUe3RlbGVtZXRyeV9pc19ub3RfZW5jcnlwdGVkfQ==`. Running **From Base64** in CyberChef immediately turns it back into **CAT**{**telemetry_is_not_encrypted**}. *How it helps beginners:* Explains the difference between encoding (which needs no key to reverse) and real encryption.

- *Dotted Line Divider (Start of 5-Stage Cryptographic Chain)*

- **Q3: The Whistleblower's Hex Packet (Chain Stage 1):** Players convert a hexadecimal dump (`43 41 54 7b ...`) using CyberChef's **From Hex** block to get **CAT**{**whistleblower_key_is_SHADOWCORP**}. This gives them the keyword `SHADOWCORP` for Q4.

- **Q4: The Encrypted Boardroom Dispatch (Chain Stage 2):** Players enter the key SHADOWCORP from Q3 into CyberChef's **Vigenère Decode** tool on UHA{klsgor_ewbrh_gg_ggnso} to get **CAT{caesar_shift_is_seven}**. This gives the shift number 7 for Q5.

- **Q5: Unlocking the Shadow Vault (Chain Stage 3):** Using the shift of 7 from Q4, players shift JHA{ahynoa_chbsa_jlyilybz} back by 7 letters to get **CAT{target_vault_cerberus}**, revealing the vault name cerberus.

- **Q6: Cracking the CISO's Password Hash (Chain Stage 4):** The vault uses an unsalted MD5 hash: f6be3f2408481885304a362deafa168a. Pasting this hash into **CrackStation** (crackstation.net) looks it up in a huge rainbow table and cracks the password in under a second: cerberus123 (**CAT{cerberus123}**). *How it helps beginners:* Seeing a 32-character hash break instantly because someone just typed 123 at the end of a word makes students rethink their own passwords right away.

- **Q7: Neutralizing the Surveillance Master Payload (Chain Stage 5):** The final payload is locked with repeating-key XOR using the password from Q6 (cerberus123) and encoded in Base64. Stacking From Base64 and XOR (with key cerberus123) in CyberChef reveals **CAT{evilcorp_surveillance_neutralized_2026}**.

D. Web Security Arena (WebSecChallenges.js & /public/websec/) Because learning web security without real webpages feels weird, we built 7 interactive EvilCorp target webpages in /public/websec/ that open right in a new browser tab.

- **Q1: The Forgotten Developer Note (Warm-Up 1):** Players open /websec/q1_login.html, right-click the page, and click **View Page Source** (Ctrl+U). At the bottom of the HTML code is a developer comment holding **CAT{always_view_the_source}**. *How it helps beginners:* Shows that anything left inside HTML comments is sent to every visitor's browser.

- **Q2: Where Search Spiders Fear to Tread (Warm-Up 2):** Players open /websec/q2_robots.txt to read the file where EvilCorp told search engines which secret folders not to list, finding **CAT{robots_reveal_secrets}**. *How it helps beginners:* Shows that robots.txt is a public text file anyone can read, not a security wall.

- *Dotted Line Divider (Start of 5-Stage Web Infiltration Chain)*

- **Q3: The Greyed-Out Export Button (Chain Stage 1):** On /websec/q3_helpdesk.html, the export button is locked on screen with the HTML disabled attribute. Players right-click the button, open **Inspect Element** (F12), delete the word disabled, and click the button to get **CAT{never_trust_the_client}** along with the cookie role admin needed for Q4.

- **Q4: The VIP Cookie Jar (Chain Stage 2):** On /websec/q4_dashboard.html, the page checks a plain browser cookie named evilcorp_role. Players change its value from

guest to admin to unlock the executive panel, getting **CAT{cookies_are_not_secure}** and finding the CEO's Employee ID (1042) for Q5.

- **Q5: Stepping Into the CEO's Shoes (IDOR, Chain Stage 3):** Players open /websec/q5_profile.html?id=1001 and change the number in the address bar from 1001 to 1042 (from Q4) to view the CEO's private profile, getting **CAT{idor_exposes_everyone}** and the server file path ../../etc/evilcorp_shadow.conf for Q6.
- **Q6: Climbing Out of the Web Folder (Path Traversal, Chain Stage 4):** On /websec/q6_files.html?file=welcome.txt, players enter ../../etc/evilcorp_shadow.conf (from Q5) to step out of the public folder and read the server's config file, getting **CAT{dot_dot_slash_escape}** and the database admin username director_vance for Q7.
- **Q7: Breaking the Master Vault with One Quote (SQL Injection, Chain Stage 5):** On /websec/q7_vault.html, players enter director_vance (from Q6) and type ' OR 1=1 -- into the password box while watching the live SQL query preview turn green, unlocking the final flag **CAT{sqli_shatters_evilcorp}**.

3. Conclusion

Working on **Cyber Awareness Academy** taught our team a lot, not just about building a full-stack web app, but also about how regular people outside our branch look at technology. What started as a simple college project turned into a full platform that helps beginners get past their fear of cybersecurity.

3.1 Key Learnings

On the coding side, we learned how to connect a React 19 frontend to a Node.js API server and set up an embedded SQLite3 database (`ctf.db`) so that it creates and fills its own tables automatically whenever someone runs `node server.js`. We also learned how to keep React state in sync across pages so that solving a flag inside an arena updates the progress bar on the main dashboard right away. On the design side, getting rid of our early rounded boxes and fake macOS window dots in favor of a clean Gruvbox Dark terminal style showed us that keeping the screen simple makes it much easier for users to focus. Most of all, seeing our survey results from 130 students and watching beginners like our persona Aditi try out our 2+5 chains proved our main point: when you remove setup headaches and put a clear, plain-English guide right next to the puzzle, anyone can learn how to protect their digital privacy.

3.2 Future Scope and Improvements

While our current offline setup works smoothly on a local laptop, there are a few upgrades we want to build next:

1. **Hosting on Our Own Server:** Right now, you run `node server.js` and `npm start` locally. Our next step is to put the app inside a Docker container and host it on our own Linux server with Nginx so anyone can play through a normal web link.

2. **Sandboxed Backend Containers for Web Security:** Our 7 EvilCorp webpages in `/public/websec/` currently simulate server behavior safely in the browser so beginners cannot break their own laptops. Next, we want to run isolated Docker backend containers so players can test real server-side SQLite queries and network headers.

3. **Built-In Mini Decoders:** Adding a small Base64/Hex/ROT13 decoder and metadata viewer right inside our React page would let beginners test quick clues without even switching browser tabs.

4. **Unique Flags Per Student:** Generating slightly different flags for each user session will let college professors use our platform for lab quizzes without students copying flags from each other.

3.3 Open-Source Sources and Inspirations

Our project was inspired by several open-source projects and free tools that make learning security possible for everyone. We took inspiration from **GCHQ's CyberChef** for our browser decoding steps, **Carnegie Mellon's picoCTF** and **OverTheWire** for our connected puzzle chains, and **Pavel Pertsev's Gruvbox** color theme for our terminal look. We also rely on free browser tools like Photopea, HexEd.it, CrackStation, Exif.tools, and the Academo Spectrum Analyzer, along with React and SQLite.

References

J. Vykopal, V. Švábenský, and E. Chang, "Benefits and pitfalls of using capture the flag games in university courses," in *Proc. 51st ACM Technical Symp. Computer Science Education (SIGCSE)*, Portland, OR, USA, 2020, pp. 752–758.

P. Chapman, J. Burket, and D. Brumley, "picoCTF: A game-based computer security competition for high school students," in *Proc. USENIX Summit Gaming, Games, and Gamification in Security Education (3GSE)*, San Diego, CA, USA, 2014, pp. 1–10.

OverTheWire Security Community, "OverTheWire: Bandit and Natas Wargames," 2024. [Online]. Available: https://overthewire.org/wargames/

TryHackMe, "Hands-on cyber security training through browser-based labs," 2024. [Online]. Available: https://tryhackme.com/

Hack The Box, "Cybersecurity training and penetration testing labs," 2024. [Online]. Available: https://www.hackthebox.com/

V. Švábenský, J. Vykopal, M. Cermak, and M. Laštovička, "Enhancing cybersecurity skills by creating serious games," in *Proc. 23rd Annu. ACM Conf. Innovation and Technology in Computer Science Education (ITiCSE)*, Larnaca, Cyprus, 2018, pp. 194–199.

S. K. Chng, H. Lu, A. Kumar, and D. Yau, "Hacker types, motivations and strategies: A comprehensive framework," *Computers in Human Behavior Reports*, vol. 5, p. 100167, Mar. 2022.

Government Communications Headquarters (GCHQ), "CyberChef: The Cyber Swiss Army Knife," GitHub Repository, 2024. [Online]. Available: https://github.com/gchq/CyberChef

P. Pertsev, "Gruvbox: Retro groove color scheme for Vim and terminal emulators," GitHub Repository, 2012. [Online]. Available: https://github.com/morhetz/gruvbox

A. Hern, "Fitness tracking app Strava gives away location of secret US army bases," *The Guardian*, Jan. 28, 2018. [Online]. Available: https://www.theguardian.com/world/2018/jan/28/fitness-tracking-app-gives-away-location-of-secret-us-army-bases

Open Web Application Security Project (OWASP), "OWASP Top 10: The Ten Most Critical Web Application Security Risks," 2021. [Online]. Available: https://owasp.org/www-project-top-ten/

S. Hetzl and P. Mutzel, "A graph-theoretic approach to steganography," in *Communications and Multimedia Security (CMS)*, Lecture Notes in Computer Science, vol. 3677. Berlin, Germany: Springer, 2005, pp. 119–128.